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## Printing apparatus and roll holding device

### Abstract


Provided is a printing apparatus for performing printing on a sheet drawn out from a roll including: a roll holding unit configured to hold the roll; a support unit configured to rotatably support the roll holding unit; a core portion provided to the roll holding unit and configured to be inserted into a hollow portion of a core pipe of the roll from an end portion of the core pipe to extend along an axis of the core pipe; a contact member provided to the core portion and configured to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and a cam member disposed inside the core portion and configured to pivot to a first position where the contact member is pressed onto the inner wall and to a second position where the contact member is retracted from the inner wall.



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## Field of Classification Search

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## Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present disclosure relates to a printing apparatus and a roll holding device and, to be specific, relates to a structure of a roll holding device configured to hold a printing medium in the shape of a roll.
Description of the Related Art
(2) As this type of a roll holding device, Japanese Patent No. 5553570 (hereinafter, referred to as PTL 1) describes a device that holds roll paper at two ends of a core pipe around which the roll paper is rolled up by fitting portions protruding from flanges to a hollow portion of the core pipe. According to this holding device, even in a case where the roll paper width is great, a wide working space is unnecessary for mounting of the roll paper, and it is possible to improve the workability. Additionally, since the holding device can be formed of the flanges and the fitting portions protruding therefrom, it is possible to relatively save the weight of the holding device.
(3) In such a holding device described in PTL 1, a leaf spring is provided to each of the above-described fitting portions inserted into the roll paper core pipe, and the roll paper is held by the elastic force of this leaf spring. For this reason, the elastic force of the leaf spring may be a relatively great resistance to the insertion at the time of inserting the fitting portion of the flange into the roll paper core pipe. Particularly, the inner diameter of the core pipe may be relatively small due to the size variation of the roll paper depending on the roll paper to be used, and accordingly the insertion resistance is increased.
SUMMARY OF THE INVENTION
(4) An aspect of the present disclosure is a printing apparatus that performs printing on a sheet drawn out from a roll in which the sheet is rolled up around a core pipe, comprising: a roll holding unit configured to hold the roll; a support unit configured to be able to rotatably support the roll holding unit; a core portion provided to the roll holding unit and configured to be inserted into a hollow portion of the core pipe from an end portion of the core pipe so as to extend along an axis of

the core pipe; a contact member provided to the core portion and configured to be able to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and a cam member disposed inside the core portion and configured to be able to pivot to a first position in which the contact member is pressed onto the inner wall and to a second position in which the contact member is retracted from the inner wall.
(5) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is an external perspective view schematically illustrating a printing apparatus according to one embodiment of the present disclosure;
(2) FIG. **2** is a schematic cross-sectional view of the printing apparatus illustrated in FIG. **1** viewed along an X direction in FIG. **2**;
(3) FIG. **3** is a block diagram illustrating a control configuration of the printing apparatus illustrated in FIGS. **1** and **2**;
(4) FIG. **4** is a perspective view describing mounting of a roll paper unit on the printing apparatus according to one embodiment of the present disclosure;
(5) FIGS. **5**A and **5**B are perspective views describing attaching of flange units to roll paper rolled up around a core pipe according to one embodiment of the present disclosure;
(6) FIGS. **6**A and **6**B are perspective views illustrating fitting members according to the embodiment of the present disclosure;
(7) FIGS. **7**A and **7**B are diagrams describing mounting of the roll paper unit on the printing apparatus according to one embodiment of the present disclosure;
(8) FIGS. **8**A to **8**C are perspective views describing position setting of a bearing member at the time of mounting the roll paper unit according to one embodiment of the present disclosure;
(9) FIG. **9** is a perspective view describing mounting of the roll paper unit on a conveyance-reference side according to one embodiment of the present disclosure;
(10) FIGS. **10**A and **10**B are perspective views describing mounting of the roll paper unit with the smallest roll paper width according to one embodiment of the present disclosure;
(11) FIGS. **11**A to **11**C are diagrams describing action of the fitting members according to a comparative example;
(12) FIG. **12** is a diagram describing action of the fitting members according to one embodiment of the present disclosure;
(13) FIG. **13** is a diagram describing action of the fitting members according to one embodiment of the present disclosure with parameters such as a distance between the fitting members;
(14) FIG. **14** is a perspective view illustrating a flange unit according to another embodiment of the present disclosure;
(15) FIG. **15** is a perspective view illustrating details of a cam member **160** illustrated in FIG. **14**;
(16) FIGS. **16**A to **16**D are diagrams describing an engagement relationship between the cam member and the flange unit and pressing and retraction of the fitting members by pivoting of the cam member according to the other embodiment of the present disclosure;
(17) FIGS. **17**A to **17**D are diagrams describing a configuration related to elastic deformation of the fitting members according to the other embodiment of the present disclosure;
(18) FIGS. **18**A and **18**B are diagrams describing the shape and action of the interference portions of the fitting member according to another additional embodiment of the present disclosure;
(19) FIG. **19** is a diagram illustrating details of a corner portion in the interference portion illustrated in FIG. **18**; and

(20) FIG. **20** is a cross-sectional view illustrating a structure of the flange unit according to the other additional embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

(21) Embodiments of the present disclosure are described below in details with reference to the drawings. Note that, the same reference numerals between the drawings indicate the same or corresponding elements.

Embodiment 1

(22) FIG. **1** is an external perspective view schematically illustrating a printing apparatus according to one embodiment of the present disclosure. A printing apparatus **2** has a form in which the elements of the printing apparatus **2** are covered with an exterior case and a cover. An operation panel **6** is provided to the exterior case on the device front surface side (a Y direction in FIG. **1**), and a user can perform various settings, input commands, and confirm information on the printing apparatus **2** through this panel, for example. A roll paper cover **50** is provided at the rear of the device so as to be opened and closed by using a shaft at a device rear end as a pivot shaft. The user can mount and unmount a roll paper unit **201**, that is a roll-shaped printing medium, on and from a roll paper setting unit of the printing apparatus **2** while the cover **50** is opened. Additionally, it is possible to prevent the roll paper unit **201** from being covered with dust and trash by closing the cover **50**. As described later in details in FIG. **8**, the printing apparatus **2** of the present embodiment can use roll paper portions with multiple roll paper widths, and the roll paper setting unit has corresponding mounting configurations. Two upper covers **24** and **25** that are opened and closed with a pivot shaft on the rear side are provided at the front of the device. It is possible to perform an operation to set the roll paper to a conveyance path, device maintenance for solving paper jam and the like, an attachment and detachment operation for ink tank replacement, and the like while the upper covers **24** and **25** are opened.

(23) FIG. **2** is a schematic cross-sectional view of the printing apparatus illustrated in FIG. **1** viewed from an X direction in FIG. **2**.

(24) The printing apparatus **2** of the present embodiment can use roll paper **201**S using the roll paper unit **201** and cut paper (not illustrated) as a printing medium (a sheet). The roll paper unit **201** is formed by rolling up the roll paper **201**S around a hollow core pipe and attaching thereto flanges for holding the roll, which are described later in FIG. **5** and followings, on two sides of the core pipe. In a case where the roll paper **201**S is used as the printing medium, the roll paper unit **201** is held by the device main body side by means of a conveyance-reference side flange sliding shaft portion **122** and a non-conveyance-reference side flange sliding shaft portion **126** (see FIGS. **5**A and **5**B) of the unit **201**. Specifically, the roll paper unit **201** is rotatably held by a conveyance-reference side flange bearing **71** and a non-conveyance-reference side flange bearing **73** (see FIGS. **7**A and **7**B) of the device main body. The roll paper **201**S drawn by the user from the roll paper unit **201** held by the bearings **71** and **73** is conveyed to the downstream side through a sheet conveyance path along a lower guide **8**. Once a leading end of the roll paper **201**S reaches a nip portion between a conveyance roller **9** and a driven roller **10**, the conveyance roller **9** is driven by a conveyance motor **51** (see FIG. **3**) to rotate, and accordingly the roll paper **201**S is nipped by the conveyance roller **9** and the driven roller **10**. Additionally, the roll paper **201**S is conveyed onto a platen **11** facing a printing head **13** by driving the conveyance roller **9** to rotate in the same state. On the other hand, in a case where the cut paper is used, a folded tray (not illustrated) is unfolded and arrayed on the lower guide **8** while the roll paper unit **201** is detached from the device **2**. The cut sheet is disposed on the tray in this state such that a leading end reaches the nip portion between the conveyance roller **9** and the driven roller **10**.

(25) The platen **11** forms an additional conveyance path for the conveyed roll paper **201**S (or the cut paper; the same applies to the description of FIG. **2** below). Also, the platen **11** supports the roll paper from the back side and maintains a constant distance between the printing head **13** and the roll paper **201**S. The printing head **13** is built in a carriage **12**, and the carriage **12** is guided and

supported by a carriage shaft **14** as a scanning guide extending in the X direction to move reciprocally in ±X directions (main scanning directions) along the scanning guide. The printing head **13** includes ejection port arrays in which multiple ejection ports (nozzles) ejecting inks of yellow (Y), cyan (C), magenta (M), and black (Bk) are arrayed in the Y direction, for example. The inks are ejected from those ejection ports of the ejection port arrays in a downward direction (a-Z direction) in accordance with image data while the carriage **12** is moving. Once an image of one line is printed by the ejection operation of the printing head **13** and the movement of the carriage **12** as described above, the roll paper **201**S is conveyed by a predetermined amount in the conveyance direction Y by the conveyance roller **9** and the driven roller **10**, and the carriage **12** is moved again to print the next line. An image is printed on the entirety of a predetermined page by repeating those operations. Ink tanks respectively corresponding to the inks Y, M, C, and Bk ejected from the printing head **13** are provided to a right end (an end portion in the −X direction in FIG. **1**) of the device **2** in the movement direction of the carriage **12**. The printing head **13** is supplied with the inks from those ink tanks through corresponding tubes. A cutter **16** is provided to the roll paper conveyance path at a predetermined distance on the downstream side of the printing head **13** and can cut the roll paper **5201** into a predetermined length. Note that, the printing operation in a case of using the cut paper as the printing medium is similar to that in the case of using the roll paper **201**S. The cut roll paper **201**S after printing ends is conveyed to a paper discharging tray **18** (see FIG. **1**).
(26) FIG. **3** is a block diagram illustrating a control configuration of the printing apparatus illustrated in FIGS. **1** and **2**.
(27) The printing apparatus **2** of the present embodiment includes a control unit **402**, and the control unit **402** controls the printing operation and the like based on a printing command transmitted from a host device **401**. Specifically, the host device **401** transmits the printing command of an image generated by the host device **401** to a main control unit **403** of the printing apparatus **2**. The control unit **402** includes the main control unit **403** and an image printing control unit **404** as a main configuration. The main control unit **403** includes a CPU **406**, a ROM **407**, and a RAM **408**, and the CPU **406** controls overall the printing apparatus **2** according to various programs and parameters stored in the ROM **407** by using the RAM **408** as a working area. The image printing control unit **404** controls driving of the conveyance motor **51**, a carriage motor **54**, the printing head **13**, an actuator of the cutter **16**, a roll paper motor **55**, and the like under instruction of the main control unit **403**.
(28) Note that, in addition to the above-described form, the form of the printing apparatus includes a printing unit in a copier or a facsimile, for example. Additionally, a multifunction peripheral to which other functions such as a printer function, a facsimile function, an original document reading function, and an imaging function are added and also a system device formed by connecting the printing apparatus to a host device such as a computer are included in the printing apparatus. It is obvious from the following descriptions that it is possible to apply the present disclosure to those devices, and in this regard, those devices are included in the printing apparatus.
(29) [Mounting of Roll Paper Unit]
(30) FIG. **4** is a perspective view describing mounting of the roll paper unit **201** on the printing apparatus **2** according to one embodiment of the present disclosure. At the time of mounting the roll paper unit **201**, flange units (described later) are attached to two ends of a core pipe **202** around which the roll paper **201**S is rolled up. This core pipe **202** may be made of paper such as cardboard. The roll paper unit **201** to which the flange units are attached as described above can be inserted and mounted on a roll paper setting unit while the roll paper cover **50** of the printing apparatus **2** is opened by the user. As illustrated in FIGS. **7**A and **7**B, the sliding shaft portions **122** and **126** of the flange units that function as a roll holding member (a roll holding device) in this process are supported by the conveyance-reference side flange bearing **71** and the non-conveyance-reference side flange bearing **73** of the roll paper setting unit. Here, the conveyance-reference side flange

bearing **71** and the non-conveyance-reference side flange bearing **73** function as a support unit.

(31) [Attaching of Flange Units to Roll Paper Portion]

(32) FIGS. **5**A and **5**B are perspective views describing attaching of the flange units into the roll paper **201**S rolled up around the core pipe **202**.

(33) The flange units function as a roll holding unit (a roll holding device) and include a conveyance-reference side flange unit **123** and a non-conveyance-reference side flange unit **124**. The flange units **123** and **124** are respectively inserted and attached to the corresponding two ends of the hollow portion of the roll paper core pipe **202**. To be specific, as illustrated in FIG. **5**A, the conveyance-reference side flange unit **123** includes a conveyance-reference side flange **131** and a conveyance-reference side flange core portion **132** protruding from one surface of the conveyance-reference side flange **131**. Additionally, the conveyance-reference side flange unit **123** includes the conveyance-reference side flange sliding shaft portion **122** protruding from a surface of the conveyance-reference side flange **131** on the opposite side of the above-described one surface and a flange gear **125** continuous from the conveyance-reference side flange sliding shaft portion **122**. Here, the surface of the conveyance-reference side flange **131** on the conveyance-reference side flange core portion **132** side is a flat surface, and with this surface, the conveyance-reference side flange **131** can be abutted (particularly, closely abutted) onto an end portion of the roll paper core pipe **202** and an end surface of the roll paper **201**S rolled up around the roll paper core pipe **202**. The conveyance-reference side flange core portion **132** includes first fitting members **150** and second fitting members **137** disposed at a predetermined distance from each other in a longitudinal direction thereof. As described later in details in FIG. **12** and the like, those first fitting member **136** and second fitting member **137** can suppress tilting of the flange unit attached to the roll paper unit **201** due to the weight of the roll paper **201**S.

(34) As with the conveyance-reference side flange unit **123**, the non-conveyance-reference side flange unit **124** includes a non-conveyance-reference side flange **135** and a non-conveyance-reference side flange core portion **134** protruding from one surface of the non-conveyance-reference side flange **135**. Additionally, the non-conveyance-reference side flange unit **124** includes a non-conveyance-reference side flange sliding shaft portion **126** protruding from the other surface of the non-conveyance-reference side flange **135**. Here, the surface of the non-conveyance-reference side flange **135** on the non-conveyance-reference side flange core portion **134** side is a convexly curved surface. For this reason, the surface can be abutted onto the roll paper core pipe **202** and the end surface of the roll paper **201**S rolled up around the roll paper core pipe **202** and can bias the roll paper unit **201** to the conveyance-reference side. As with the conveyance-reference side, the non-conveyance-reference side flange core portion **134** includes first fitting members **138** and second fitting members **139** disposed at a predetermined distance from each other. As described later in details, those first fitting members **138** and second fitting members **139** suppress tilting of the flange unit due to the weight of the roll paper **201**S.

(35) In the above-described configuration, the pair of the fitting member **136** on the conveyance-reference side are disposed at intervals of 180 degrees along a circumference of the flange core portion **132**. The pair of the fitting member **137** on the conveyance-reference side are also disposed at intervals of 180 degrees along a circumference of the flange core portion **132**. Similarly, the pair of the fitting member **138** on the non-conveyance-reference side are disposed at intervals of 180 degrees along a circumference of the flange core portion **134**. The pair of the fitting member **139** on the non-conveyance-reference side are also disposed at intervals of 180 degrees along a circumference of the flange core portion **134**. One of the pair of fitting members **136** and one of the pair of the fitting member **137** are disposed at the same positions along circumferences of the flange core portion **132**. This also applies to the other of the pair of fitting member **136** and the other of the pair of the fitting member **137**. Similarly, one of the pair of fitting members **138** and one of the pair of the fitting member **139** are disposed at the same positions along circumferences of the flange core portion **134**. This also applies to the other of the pair of fitting member **138** and

the other of the pair of the fitting member **139**. As a result, each of flange core portions **132**, **134** includes the four fitting members. Note that, the number of the fitting members is not limited thereto and can be determined in accordance with specifications of the device and the like. Likewise, it is not limited to the mode that the number of the fitting members **137**, **139** respectively close to the flange **131**, **135** and the number of the fitting members **136**, **138** respectively far from the flange **131**, **135** in their respective flange core portion are equal to each other.

(36) FIGS. **6**A and **6**B are perspective views illustrating the fitting members according to the embodiment of the present disclosure. FIG. **6**A illustrates the first fitting member **136**, **138** and FIG. **6**B illustrates the second fitting member **137**, **139**. The fitting member **136**, **138** includes a flat surface portion **152** and a pair of interference portions **153** which form a pair of wings. A metallic plate member is bended along a boundary between the flat surface portion **152** and one of the pair of interference portion **153** and along another boundary between the flat surface portion **152** and the other of the pair of interference portion **153**. Similarly, the fitting member **137**, **139** includes a flat surface portion **156** and a pair of interference portions **157** which form a pair of wings. A metallic plate member is bended along a boundary between the flat surface portion **156** and one of the pair of interference portion **157** and along another boundary between the flat surface portion **156** and the other of the pair of interference portion **157**. With this bending, the fitting member **136**, **138** has the elasticity against the opposite forces acting on the flat surface portion **152** and the pair of interference portions **153**. This also applies to the fitting member **137**, **139**. The fitting members **136**, **138** are respectively bent from the flat surface portions **152** at constricted portions **151**, and the bent portion functions as a fixing portion **150** screwed on the flange unit. The fitting members **137**, **139** are respectively bent from the flat surface portions **156** at constricted portions **155**, and the bent portion functions as a fixing portion **154** screwed on the flange unit.

(37) As illustrated in details in FIG. **14**, the fixing portion **154** of the fitting member **137** is screwed on the flange **131**. Likewise, as illustrated in detail in FIG. **14**, the above-described fixing portion **150** of the fitting member **136** is screwed on an end surface of the flange core portion **132**. Similarly, although no figure shows, the fixing portion **154** of the fitting member **139** is screwed on the flange **135**. Likewise, the above-described fixing portion **150** of the fitting member **138** is screwed on an end surface of the flange core portion **134**. In the present embodiment, with respect to each of fitting members **136**, **137**, **138**, **139**, the fixing portion **150** or **154** is fixed by a screw as described above, and the flat surface portion **152** or **156** is in contact with an outer surface of the corresponding flange core portion **132**. Note that, an opening may be provided in a region of the flange core portion **132** corresponding to portions other than the fixing portion **150** or **154** of the fitting member, and the fitting member is deformed so as to enter the inside of a hollow portion of the flange core portion through this opening. In this case, it is favorable that the elastic force of the fitting member is relatively great enough so that the core pipe **202** of the roll paper unit **201** can be appropriately pressed and held by the elastic force.

(38) In order to attach the conveyance-reference side flange unit **123** and the non-conveyance-reference side flange unit **124** to the roll paper portion, first, each of the conveyance-reference side flange core portion **132** and the non-conveyance-reference side flange core portion **134** is inserted into the hollow portion of the core pipe **202** of the roll paper portion. The outer diameters of both of the conveyance-reference side flange core portion **132** and the non-conveyance-reference side flange core portion **134** except for the portions of the fitting members **136** and **138** are designed smaller than the inner diameter of the hollow portion of the core pipe **202** of the roll paper portion. On the other hand, each of the first fitting members **136**, the second fitting members **137**, the first fitting members **138**, and the second fitting members **139** has the shape of a leaf spring that will be elastically deformed as described above in FIG. **6**. Thus, those fitting members **136-139** can have a dimension relationship for being fitted with the hollow portion of the roll paper core pipe **202**, in other words, a dimension relationship for being in contact with the roll paper core pipe **202** with force acting on each other.

(39) FIG. **5**B illustrates a state where the conveyance-reference side flange unit **123** and the non-conveyance-reference side flange unit **124** are attached to the roll paper portion including the core pipe **202** and the roll paper rolled up around the core pipe **202**. Hereinafter, a unit in this attached state is also referred to as a “roll unit”.

(40) [Mounting of Roll Paper Unit on Device Main Body]

(41) FIGS. **7**A and **7**B are diagrams describing mounting of the roll paper unit on the printing apparatus **2**. FIG. **7**A illustrates a state before the roll paper unit, in other words, the roll paper portion to which the conveyance-reference side flange unit **123** and the non-conveyance-reference side flange unit **124** are attached, is mounted on the printing apparatus **2**, and FIG. **7**B illustrates a state where the roll paper unit is mounted on the printing apparatus **2**.

(42) As illustrated in FIG. **7**A, the conveyance-reference side flange sliding shaft portion **122** is set to the conveyance-reference side flange bearing **71**, and the non-conveyance-reference side flange sliding shaft portion **126** is set to the non-conveyance-reference side flange bearing **73**. In a case where the user performs this mounting, it is possible to make positioning so as to fit the conveyance-reference side flange sliding shaft portion **122** to the conveyance-reference side flange bearing **71** and the non-conveyance-reference side flange sliding shaft portion **126** to the non-conveyance-reference side flange bearing **73**, respectively. Particularly, even in a case where the conveyance-reference side flange sliding shaft portion **122** is close to the conveyance-reference side flange bearing **71** and the non-conveyance-reference side flange sliding shaft portion **126** is close to the non-conveyance-reference side flange bearing **73**, respectively, the operation can be performed while the user can see the respective fitting positions. Thus, it is possible to facilitate positioning at the time of mounting the roll paper unit **201** on the printing apparatus **2**. As also illustrated in FIG. **7**B, even in a state after the mounting of the roll paper unit **201** on the printing apparatus **2** is completed, the conveyance-reference side flange sliding shaft portion **122** and the conveyance-reference side flange bearing **71** can be investigated visually. Also, the non-conveyance-reference side flange sliding shaft portion **126** and the non-conveyance-reference side flange bearing **73** can be investigated visually. Accordingly, it is obvious that those constituents can be investigated visually even in the middle of mounting.

(43) [Position Setting of Bearing Member]

(44) FIGS. **8**A, **8**B, and **8**C are perspective views from obliquely above describing position setting of the bearing member in a case of mounting the roll paper unit and illustrate the roll paper setting unit of the printing apparatus **2** with the opened roll paper cover **50**.

(45) FIG. **8**A illustrates a state where no bearing portion is set to a non-conveyance-reference side disposing portion of the roll paper unit **201** in the printing apparatus **2**. FIG. **8**B illustrates a state where a non-conveyance-reference side bearing member **180** for mounting the roll paper unit **201** with a wide roll paper width on the non-conveyance-reference side disposing portion is set (attached). FIG. **8**C illustrates a state where the non-conveyance-reference side bearing member **180** for mounting the roll paper unit **201** with a narrow roll paper width on the non-conveyance-reference side disposing portion is set.

(46) In FIG. **8**A, at the time of mounting the roll paper unit **201** on the printing apparatus **2**, a positioning projection **182** of the non-conveyance-reference side bearing member **180** is inserted into any one of positioning holes **184**, **185**, and **186** corresponding to the roll paper widths in the non-conveyance-reference side disposing portion. The positioning holes **184**, **185**, and **186** are set in accordance with the three sizes of the roll paper widths assumed to be used in the printing apparatus **2**. Note that, the positioning holes provided at three portions are indicated in the example illustrated in FIG. **8**A; however, as a matter of course, positioning holes may be provided at any number of positions corresponding to the number of sizes of the roll paper width assumed to be used. In the present embodiment, the positioning hole **184** corresponding to the maximum roll paper width of the roll paper unit **201**, the positioning hole **185** corresponding to a smaller roll paper width than the above, and the positioning hole **186** corresponding to a further smaller roll

paper width than the above are provided. FIG. **8**B illustrates a state where the non-conveyance-reference side bearing member **180** is attached to the positioning hole **184** corresponding to the greatest roll paper width. FIG. **8**C illustrates a state where the non-conveyance-reference side bearing member **180** is attached to the positioning hole **186** corresponding to the smallest roll paper width.

(47) FIG. **9** is a perspective view from obliquely above describing mounting of the roll paper unit on the conveyance-reference side and illustrates the roll paper setting unit on the conveyance-reference side of the printing apparatus **2** with the opened roll paper cover **50**.

(48) At the time of attaching the conveyance-reference side flange unit **123** to the printing apparatus **2**, the conveyance-reference side flange sliding shaft portion **122** (FIG. **7**) is engaged to the conveyance-reference side flange bearing **71**. In this process, the flange gear **125** (FIG. **7**) is engaged with a driving gear **72** (FIG. **9**) provided on the main body side of the printing apparatus **2**. This driving gear **72** is rotated and driven by the roll paper motor **55** (FIG. **3**) provided on the main body side of the printing apparatus **2**. Thus, the conveyance-reference side flange unit **123** can rotate in forward direction and reverse direction with the roll paper unit **201** and the non-conveyance-reference side flange unit **124**, whereby a feeding operation by the roll paper unit **201** is performed.

(49) FIGS. **10**A and **10**B are perspective views describing mounting of the roll paper unit with the smallest roll paper width. FIG. **10**A illustrates a state before the roll paper unit **201** is mounted on the printing apparatus **2**, and FIG. **10**B illustrates a state after the roll paper unit **201** is mounted on the printing apparatus **2**. As illustrated in FIG. **10**A, as with the mounting in a case where the roll paper unit **201** has a great roll paper width, the conveyance-reference side flange sliding shaft portion **122** is engaged to the conveyance-reference side flange bearing **71**, and the non-conveyance-reference side flange sliding shaft portion **126** is engaged to the non-conveyance-reference side flange bearing **73**. Accordingly, ease of mounting the roll paper unit **201** on the printing apparatus is similar to the case where the roll paper width is wide. As illustrated in FIG. **10**B, even after the roll paper unit **201** is mounted on the printing apparatus **2**, the conveyance-reference side flange sliding shaft portion **122** and the conveyance-reference side flange bearing **71** can be investigated visually. Also, the non-conveyance-reference side flange sliding shaft portion **126** and the non-conveyance-reference side flange bearing **73** can be investigated visually similarly to the case of mounting of the roll paper unit **201** with a wide paper width, and mounting of the roll paper unit **201** on the printing apparatus is facilitated.

(50) [Action of Fitting Members in Flange Unit]

(51) The fitting members according to one embodiment of the present disclosure suppress each flange unit from tilting due to the weight of the roll paper portion. FIGS. **11**A to **11**C are diagrams describing action of the fitting members according to a comparative example, and FIG. **12** is a diagram describing action of the fitting members according to one embodiment of the present disclosure. Any of FIGS. **11**A to **11**C and **12** illustrate a state where the roll paper unit **201** formed by attaching the flange unit to the roll paper portion is mounted on the printing apparatus **2**.

(52) As illustrated in FIG. **11**A, a flange unit **701** according to the comparative example includes the pair of the fitting members **137** on the flange core portion **132**. The conveyance-reference side flange sliding shaft portion **122** of the flange unit **701** is slidably supported by the conveyance-reference side flange bearing **71** on the device main body side. In this case, the entirety of the flange unit **701** may tilt as illustrated in FIG. **11**B depending on the weight of the roll paper unit **201**. If the roll paper is fed from the roll paper unit **201** in such a state, inconvenience occurs such as displacement of an end portion position of the fed roll paper. To deal with this, as illustrated in FIG. **11**C, there may be considered a configuration in which a shaft portion **702** extending from the flange unit **701** is provided, and also a member **700** that applies acting force in the opposite direction of the tilting direction to the shaft portion **702** is provided on the printing apparatus **2** side. However, in a case of this configuration, there are needed a mechanism that retracts the

member **700** from the extending shaft portion **702** of the flange unit at the time of attaching the flange unit, a mechanism that moves the member **700** to a pressing position after the flange unit is attached, and the like. For this reason, there is inconvenience that the configuration of the main body device side is complicated, and the cost is increased.

(53) To deal with this, in one embodiment the present disclosure, fitting members **137***a* and **137***b* (second contact members) and fitting members **136***a* and **136***b* (first contact members) apart from each other by a predetermined distance are provided to the conveyance-reference side flange core portion **132** as illustrated in FIG. **12**. Thus, each of the fitting members **137***a* and **137***b* and the fitting members **136***a* and **136***b* is put in contact with the core pipe **202** of the roll paper unit **201** with no clearance. Additionally, the distance between the fitting members **137***a*, **137***b* and the fitting members **136***a*, **136***b* causes the moment against the moment causing the flange unit to tilt, and thus the tilting can be suppressed.

(54) Note that, the fitting members **136** and the fitting members **137** may be provided to be distributed over the entire circumference of the inner wall surrounding the hollow portion of the core pipe **202**. However, in this configuration, the cost is increased, and the force required to insert the conveyance-reference side flange unit **123** into the core pipe **202** of the roll paper portion is increased. For this reason, in the present embodiment, the fitting portions are not provided over the entire circumference, but instead the fitting members **136***a* and **136***b* are provided at two portions at an interval of 180 degrees in the circumferential direction. Likewise, the fitting members **137***a* and **137***b* are provided at two portions at an interval of 180 degrees in the circumferential direction. Here, each of the fitting members **136***a* and **136***b* has the two interference portions **153** in the shape of wings, and each of the fitting members **137***a* and **137***b* has the two interference portions interference portions **157** in the shape of wings. Accordingly, the four interference portions **153** and the four interference portions **157** in total are put in contact with the inner wall surrounding the hollow portion of the core pipe **202**. Accordingly, regarding the conveyance-reference side flange **123**, the total number of the interference portions (the interference portions **153** and the interference portions **157**) put in contact with the inner wall surrounding the hollow portion of the core pipe **202** is eight.

(55) As above, the situation described above with reference to FIGS. **11**A to **11**C and **12** is similarly applied to the flange unit on the non-conveyance-reference side. That is, regarding the non-conveyance-reference side flange **124**, each of the fitting members **138***a* and **138***b* includes the two interference portions **153** in the shape of wings, and each of the fitting members **139***a* and **139***b* includes the two interference portions **157** in the shape of wings. Accordingly, regarding the non-conveyance-reference side flange **124**, the total number of the interference portions (the interference portions **153** and the interference portions **157**) put in contact with the inner wall surrounding the hollow portion of the core pipe **202** is eight.

(56) FIG. **13** is a diagram describing action of the fitting members according to one embodiment of the present disclosure with parameters such as a distance between the fitting members. In considering the tilting of the flange unit due to the weight of the roll paper, the force illustrated in FIG. **13** acts on each element of the roll paper unit **201**. In other words, Mg is the weight of a combination of the core pipe **202** and the rolled up roll paper (the weight of the roll paper portion). F1 is reaction force that acts on each fitting member **136** (a point a) from the inner wall surrounding the hollow portion of the core pipe **202**, F2 is reaction force that acts on each fitting member **137** (a point b) from the inner wall surrounding the hollow portion of the core pipe **202**, and N is reaction force that acts on the flange sliding shaft portion **122** from the flange bearing **71**. Note that, the weight Mg of the roll paper portion is different in accordance with the width of the roll paper used in the printing apparatus and is different also in accordance with the length of the roll paper rolled up around the core pipe **202**. The weight Mg of the roll paper portion indicates the maximum value in a case where the roll paper with the maximum width is unused. In the present consideration, the weight Mg of the roll paper portion in this case is used. This is for considering a

state where the tilt of the flange unit is the greatest. The point of action of force such as the points a and b is the center of a figure of a contact region (point, line, or surface) that is formed in a case where the fitting member and the sliding axis are put in contact with the core pipe and the bearing. In FIG. **13**, A is a distance between the fitting member **136** (the point a) and the fitting member **137** (the point b), and B is a distance between the fitting member **137** (the point b) and the flange bearing **71** (the point of action). Here, those distances A and B are distances along the insertion direction in a case where the flange core portion **132** is inserted into the core pipe **202** of the roll paper portion. Note that, as a matter of course, the following descriptions are given with a calculation result taking into consideration also the flange unit on the non-conveyance-reference side.

(57) The drags F1, F2, and N are obtained as functions of the weight Mg and the distances A and B using the relationship of the forces illustrated in FIG. **13**, that is, based on a combination of a force balance of the roll paper portion in the weight direction, a force balance of the flange units on the two sides in the weight direction, and a moment balance at each of the flange units on the two sides. Accordingly, the elastic deformation amount due to the above-described reaction force F1 and F2 at the fitting member **136** (the point a) and the fitting member **137** (the point b) can be obtained, and subsequently, based on this elastic deformation amount, the tilt moduli of the fitting member **136** (the point a) and the fitting member **137** (the point b) with respect to the distance A can be obtained. At last, based on the obtained drags and tilt moduli described above, the tilt modulus of the flange unit on each of the two sides can be obtained.

(58) According to the tilt modulus, it is desirable for the distance A between the first fitting member **136** and the second fitting member **137** to be equal to or greater than the distance B between the second fitting member **137** and the flange bearing **71** (the distance A is equal to or greater than the distance B). This is because the above-described tilt modulus has a characteristic to be smaller in inverse proportion to a distance ratio A/B squared. In other words, regarding this characteristic, it is confirmed in the printing apparatus of the present embodiment that the tilting of the flange unit is effectively suppressed as long as A/B is one or greater, experimentally. On the other hand, for example, the upper limit of A/B can be determined in terms of ease of handling the flange member taking into consideration the lengths of the flange core portions **132** and **134** on which the fitting members are disposed.

(59) Note that, in terms of suppressing the tilting of the flange unit, the form of the fitting members is not limited to the above-described leaf spring form. The fitting members may be any elastic body as long as that the elastic body is disposed on the flange core portion **132**, **134** at a predetermined distance in the length direction of the flange core portion **132**, **134** and is put in contact with the roll paper core pipe **202** with appropriate elasticity while the flange unit is attached thereto. The fitting members may be a rubber member, for example.

(60) In the above-described example, the first and second fitting members **136** and **137** are disposed to be arrayed on the same straight line along the longitudinal direction of the flange core portion **132**, **134**. In other words, the first and second fitting members **136** and **137** are disposed at the same position in the circumferential direction of the flange core portion **132**, **134**. However, it is not limited to this form. As long as the first fitting member **136** and the second fitting member **137** are provided away from each other at a predetermined distance along the longitudinal direction as described above, the first fitting member **136** and the second fitting member **137** may be disposed at different positions in the circumferential direction of the flange core portion.

(61) Additionally, the number of the positions at which such elastic bodies are disposed in the circumferential direction of the flange core portion (positions at which the elastic bodies are put in contact with the inner wall surrounding the hollow portion of the core pipe) is two for one fitting member and is four in total for the two fitting members in the above-described embodiment; however, it is not limited thereto. Preferably, the positions at which such elastic bodies are disposed in the circumferential direction of the flange core portion are at least three positions. It is favorable

that the three positions are arranged at intervals of 120 degrees. In other words, it is favorable that such elastic bodies are arranged at positions at three equal intervals in the circumferential direction of the flange core portion, and the elastic bodies are put in contact with the inner wall surrounding the hollow portion of the core portion at those positions. Thus, it is possible to align the centers of the flange unit and the hollow portion.

Embodiment 2

(62) FIG. **14** is a perspective view illustrating the flange unit **123** on the conveyance-reference side according to an embodiment 2 of the present disclosure. The present embodiment is different from the above-described embodiment in that the fitting members **136** and **137** on the flange core portion **132** are operated by a cam member **160** that is provided inside the flange core portion **132** to form a mechanism for operating the fitting members **136** and **137**. In other words, the cam member **160** pivots between a pressing position in which the fitting members **136** and **137** press the core pipe **202** of the roll paper and a retraction position in which the fitting members **136** and **137** is not put in contact with the core pipe **202** of the roll paper. In this regard, below the fitting members **136**, **137**, an opening is formed in a portion of the flange core portion **132**. Note that, as with the above-described embodiment, since the flange unit **124** on the non-conveyance-reference side has a similar structure, only the flange unit **123** on the conveyance-reference side is described in the descriptions below. It is obvious from the following descriptions that it is possible also in the present embodiment to suppress the tilting of the flange units **123**, **124** as with the above-described embodiment.

(63) As illustrated in FIG. **14**, inside of the flange core portion **132** is hollow, and a cam member **160** is provided in the hollow so as to be pivotable about the longitudinal direction axis thereof. FIG. **15** illustrates the cam member **160**. As illustrated in FIG. **15**, the cam member **160** includes two pressing portions **161** and two pressing portions **162**. Those pressing portions **161** and **162** correspond to the two fitting members **136** and **137** provided to the flange core portion **132**, respectively, and are put in contact with and press the flat surface portions **152** and **156** of the fitting members **136** and **137** in accordance with the pivoting of the cam member **160**. Additionally, the cam member **160** includes sliding shaft portions **160**A and **160**B in the form of a column at two ends thereof, respectively, and those sliding shaft portions **160**A and **160**B are slidably held by corresponding portions of the flange unit **123** as described later in details with reference to FIG. **16** and the like. Moreover, a lever portion **163** is provided on the circumference of the sliding shaft portion **160**B at one end. The user can operate the lever portion **163** so as to move the pressing portions **161** and **162** by pivoting the cam member **160** between the position at which the flat surface portions **152** and **156** of the fitting members **136** and **137** are pressed (the pressing position) and the position at which no pressing is performed (the retraction position). A pivoting range of the lever portion **163** for this movement between the pressing position and the retraction position is indicated by θ in FIG. **15**, and the pivoting is restricted within this range by the engagement relationship between the lever portion **163** the flange unit **123** as described with reference to FIG. **16** and the like.

(64) In the present embodiment, by pivoting, the cam member **160** can reciprocate between the pressing position in which the cam member **160** presses the fitting members **136** and **137** and the retraction position in which the cam member **160** retracts from the pressing position. Attaching the flange unit **123** to the roll paper portion while the cam member **160** is moved to the retraction position makes it possible to reduce the resistance between the fitting members **136** and **137** and the inner wall surrounding the hollow portion of the core pipe during the attaching and to improve the operation performance. Here, the retraction position is a position in which the fitting members **136** and **137** are not put in contact with the inner wall surrounding the hollow portion of the core pipe or at which the fitting members **136** and **137** are put in contact with the inner wall but are easily deformed due to the contact (by relatively small force).

(65) In other words, the inner diameter of the hollow portion of the core pipe **202** of the roll paper

unit may have a variation depending on the type and the production lot of the roll paper. Here, if the fitting members **136** and **137** are formed in accordance with the dimension of the hollow portion with a small inner diameter in the embodiment 1, the fitting members **136** and **137** cannot interfere and cannot be put in contact with the hollow portion at the time of inserting the fitting members **136** and **137** into the hollow portion with a relatively great inner diameter. For this reason, the roll paper portion cannot be held appropriately, and additionally the tilting of the flange unit **123** described above in the embodiment 1 cannot be suppressed. Therefore, in the embodiment 1, the dimension and the like of the fitting members **136** and **137** are determined in accordance with the dimension of the hollow portion with a great inner diameter. However, in this case, the fitting members **136** and **137** are deformed greatly to be inserted in a case of attaching those fitting members to the hollow portion with a relatively small inner diameter, and the force required for the attachment is increased.

(66) To deal with this, in the present embodiment, the fitting members **136** and **137** are moved to the retraction position at the time of attaching the flange unit **123** to the roll paper portion as described above. On the other hand, in a case where the flange unit **123** is attached to the roll paper portion and thereafter the roll paper unit **201** is formed to be used, the fitting members **136** and **137** are moved to the pressing position in which the fitting members **136** and **137** are put in contact with and press the hollow portion of the roll paper.

(67) Referring back to FIG. **14**. In the structure of the flange unit **123**, the fitting members **136** and **137** reciprocate between the pressing position and the retraction position by the above-described cam member **160**. For this configuration, the flange core portion **132** below the fitting members **136** and **137** is hollow. Thus, the flat surface portions **152** and **156** of the fitting members **136** and **137** and the pressing portions **161** and **162** of the cam member **160** can be put in contact with each other.

(68) The first fitting member **136** is screwed on the conveyance-reference side flange unit **123** by the fixing portion **150**, and the second fitting member **137** is screwed on the conveyance-reference side flange unit **123** by the fixing portion **154**.

(69) As illustrated in details in FIGS. **16**A and **16**B, the lever portion **163** of the cam member **160** is positioned on the back surface side of the flange **131**, in other words, a side on which the conveyance-reference side flange sliding shaft portion **122** is disposed. The flange **131** has a recess portion formed of a groove extending in the circumferential direction on the back surface side, and the lever portion **163** can be operated in the recess portion.

(70) FIGS. **16**A to **16**D are diagrams describing the engagement relationship between the cam member **160** and the flange unit **123** and the pressing and the retraction with respect to the fitting members **136** and **137** by the pivoting of the cam member **160**. Among the drawings, FIGS. **16**A and **16**B are cross-sectional views in which the conveyance-reference side flange unit **123** is viewed from a radial direction of the roll paper, and FIGS. **16**C and **16**D are cross-sectional views in which the conveyance-reference side flange unit **123** is viewed from a rotation axis direction of the roll paper unit **201**. FIGS. **16**A and **16**C illustrate orientations of the fitting members **136** and **137** and the cam member **160** in a position **1**, and FIGS. **16**B and **16**D illustrate orientations of the fitting members **136** and **137** and the cam member **160** in a position **2**.

(71) As illustrated in FIGS. **16**A and **16**B, the sliding shaft portion **160**A at one end of the cam member **160** is slidably engaged with a leading end portion of the flange core portion **132**, and the sliding shaft portion **160**B at the other end is slidably engaged with a groove bottom portion on back surface side of the flange **131**. Thus, the cam member **160** is pivotably held by the flange unit **123**, and each pivot position is maintained. The lever portion **163** of the cam member **160** is provided from the extending portion of the sliding shaft portion **160**B so as to extend perpendicularly to the extending portion. The lever portion **163** extends to the outside of a protrusion portion **131**A forming a part of the flange **131** through a slot (groove) formed on the protrusion portion **131**A, and thus the user can operate the lever portion **163**. This slot is formed on

a part of the circumferential direction of the protrusion portion **131**A, and thus the pivoting range of the lever portion **163** is restricted to the angle range θ illustrated in FIG. **15**.

(72) As illustrated in FIGS. **16**A and **16**C, the position **1** is the retraction position in which the pressing portions **161** and the pressing portions **162** of the cam member **160** do not press the flat surface portion **152** of the first fitting member **136** and the flat surface portion **156** of the second fitting member **137**, respectively. On the other hand, as illustrated in FIGS. **16**B and **16**D, the position **2** is the pressing position in which the pressing portions **161** and the pressing portions **162** press the flat surface portion **152** of the first fitting member **136** and the flat surface portion **156** of the second fitting member **137**, respectively. It is possible to switch between the position **1** and the position **2** by operating the lever portion **163** in the direction to pivot with respect to the longitudinal direction axis line of the roll paper unit **201**.

(73) FIGS. **17**A to **17**D are diagrams describing a configuration related to the deformation of the fitting members. Among the drawings, FIGS. **17**A and **17**B illustrate cross-sectional views along a radial direction of the flange unit, and FIGS. **17**C and **17**D illustrate cross-sectional views along a rotation axis direction of the flange unit. As described above, the cam member **160** is switched to the position **1** in a case of attaching the flange unit **123** to the roll paper portion. In this position **1**, the interference portion **153** of the first fitting member **136** and the interference portion **157** of the second fitting member **137** may be put in contact with the inner wall surrounding the hollow portion of the core pipe **202** of the roll paper portion. With this contact, the first fitting member **136** and the second fitting member **137** are deformed in a direction away from the core pipe **202**. In this case, as illustrated in FIG. **6**, the interference portions **153** and the fixing portion **150** are apart from each other by a predetermined distance, the interference portions **157** and the fixing portion **154** are apart from each other by a predetermined distance. The width of the constricted portion **151** and the constricted portion **155** is set small. Thus, it is possible to reduce the force required to deform the first fitting member **136** and the second fitting member **137**. Thus, it is possible to reduce the frictional force between the interference portions **153**, **157** and the inner wall surrounding the hollow portion of the roll paper core pipe **202**, and it is possible to reduce the attachment load. As a result, the operation performance at the time of attaching the flange unit **123** to the roll paper portion is improved. After the flange unit **123** is inserted into the roll paper portion, the cam member **160** is switched to the position **2** as illustrated in FIGS. **17**B and **17**D. As described above, in the position **2**, the pressing portions **161** and the pressing portions **162** press the flat surface portion **152** of the first fitting member **136** and the flat surface portion **156** of the second fitting member **137**, respectively. As a result, the interference portions **153** and the interference portions **157** are put in contact with the core pipe **202** of the roll paper, and the flange unit **123** is fixed into the core pipe **202**. The first fitting member **136** and the second fitting member **137** are elastic bodies. Accordingly, with the elastic deformation of those fitting members **136** and **137**, it is possible to suppress the displacement of the roll paper portion in which the flange unit **123** is fixed into the core pipe **202** of the roll paper portion by absorbing the variation of the inner diameter of the core pipe **202** of the roll paper.

(74) Note that, the cam member **160** includes the pressing portions **161** and **162** corresponding to the first and second fitting members **136** and **137** in the above-described embodiment; however, only the pressing portions **161** may be included corresponding to only the first fitting members **136** at the position far from the flange **131**. According to this form, a configuration in which the first fitting members **136** that are put in contact with the core pipe **202** first at the time of inserting the flange core portion **132** into the core pipe **202** of the roll paper can be retracted is obtained, and it is possible to secure the operation performance of attaching the flange unit also in this form.

(75) Additionally, the positions **1** and **2** of the fitting members **136** and **137** are switched by operating the lever portion of the cam member in the above-described embodiment 2; however, it is not limited to this form. For example, the force in a longitudinal vertical axis direction of the flange core portion **132**, **134** may act through a spring. In this case, a member that develops force in the

axis direction by the cam action into a direction crossing a longitudinal vertical axis is used. The fitting members are then pressed by the member.

Embodiment 3

(76) An embodiment 3 of the present disclosure relates to the interference portions **153** and **157** of the first fitting member **136** and the second fitting member **137**. FIGS. **18**A and **18**B are diagrams describing the shape and action of the interference portions of the fitting member. The fitting members **136** and **137** of the conveyance-reference side flange unit **123** are described below, and descriptions of the fitting members **136** and **137** of the non-conveyance-reference side flange unit **124** are omitted since they have similar configurations.

(77) As illustrated in FIG. **18**A, a corner portion **158** is provided to each interference portion **153** of the first fitting member **136**. As illustrated in FIG. **18**B, in the position **2**, the flat surface portion **152** of the first fitting member **136** is pressed by the pressing portion **161** of the cam member **160**, and the interference portion **153** is put in contact with the inner wall surrounding the hollow portion of the core pipe **202** of the roll paper portion. Thus, in a case where any force acts in a direction in which the roll paper portion is drawn from the conveyance-reference side flange unit **123** (a direction in which the roll paper portion is moved from right to left of FIG. **18**B with respect to the conveyance-reference side flange unit **123** in FIG. **18**B), the corner portions **158** dig into the wall of the core pipe **202**. Thus, it is possible to suppress the movement (displacement) of the roll paper portion with respect to the conveyance-reference side flange unit **123**. Note that, in a case of the non-conveyance-reference side, the above-described drawing direction is changed to the opposite direction.

(78) FIG. **19** is a diagram illustrating details of the corner portion. As illustrated in FIG. **19**, in a case where an angle formed by one of two crossing straight lines forming the corner portion **158** and the longitudinal direction axis of the core pipe **202** is $\alpha$, and an angle formed by the other of the two crossing straight lines is $\beta$, it is favorable that $\alpha<\beta$. Here, as illustrated in FIGS. **18** and **19**, $\alpha$ is an angle formed by the straight line farther from the flange **131** out of the two straight lines forming the corner portion **158**, and $\beta$ is an angle formed by the straight line closer from the flange **131** out of the two straight lines. In other words, $\alpha$ is an angle formed by the straight line farther from an end portion of the core portion on a rear side in the insertion direction into the core pipe, and $\beta$ is an angle formed by the straight line closer therefrom. Since the two crossing straight lines have such an angle relationship, the corner portion **158** can have a directionality to be against the direction in which the roll paper portion is drawn from the flange core portion **132**, and it is possible to suppress the movement (displacement) of the roll paper portion with respect to the conveyance-reference side flange unit **123** more successfully. An apex of the corner portion **158** has a substantially round shape, and a radius R of this rounded portion can be determined in accordance with a desired amount of the digging of the corner portion **158** into the wall of the core pipe **202**.

(79) According to the above embodiment, since the function of suppressing the displacement of the roll paper portion with respect to the conveyance-reference side flange unit **123** is added, it is possible to reduce the required pressing amount by the pressing portions **161** to the first fitting member **136**. Consequently, it is possible to further reduce the operation force required to operate the lever portion **163** from the position **1** to the position **2**.

(80) The shape of the corner portion **158** of the first fitting member **136** is described above; however, the second fitting member **137** may have a corner portion shape similar to that of the first fitting member **136** as illustrated in FIG. **20**. Additionally, as illustrated in FIG. **20**, the fixing portion **154** of the second fitting member **137** may have the same configuration as that of the fixing portion **150** of the first fitting member **136**, and a form in which a portion close to the conveyance-reference side flange **131** is pressed by the second fitting member **137** as modified may be applied.

(81) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The

scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(82) This application claims the benefit of Japanese Patent Application No. 2022-043691 filed Mar. 18, 2022, which is hereby incorporated by reference wherein in its entirety.

## Claims

1. A printing apparatus that performs printing on a sheet drawn out from a roll in which the sheet is rolled up around a core pipe, comprising: a roll holding unit including a flange and configured to hold the roll; a core portion extending from one side of the flange along a rotation axis of the roll and configured to be inserted into a hollow portion of the core pipe from an end portion of the core pipe; a shaft portion extending from the other side of the flange along the rotation axis; a flange bearing including a bearing and configured to rotatably support the shaft portion by the bearing in a case where the roll holding unit is mounted on the printing apparatus; a contact member provided to the core portion and configured to be able to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and a cam member disposed inside the core portion and configured to rotate about the rotation axis between a first position in which the contact member is pressed onto the inner wall and a second position in which the contact member is retracted from the inner wall.
2. The printing apparatus according to claim 1, wherein the cam member pivots in accordance with an operation of a lever portion attached to the cam member.
3. The printing apparatus according to claim 1, wherein the contact member is in a form of a leaf spring formed by bending a plate member at two bent portions and includes a flat surface portion and two wing-shaped portions continuously contact with the flat surface portion at a tilt through the two bent portions, and the cam member is able to press the flat surface portion.
4. The printing apparatus according to claim 3, wherein the two wing-shaped portions each include a corner portion that is put in contact with the inner wall of the core pipe.
5. The printing apparatus according to claim 4, wherein the corner portion of the wing-shaped portion has an angle formed by two crossing straight lines, and while the core portion is inserted in the core pipe, an angle formed by the inner wall and one of the two straight lines that is closer to an end portion of the core portion on a rear side in an insertion direction of the core portion into the core pipe is greater than an angle formed by the inner wall and the other one of the two straight lines.
6. The printing apparatus according to claim 1, wherein the flange is abutted onto an end portion of the core portion.
7. The printing apparatus according to claim 1, wherein the contact member includes a first contact member and a second contact member that is disposed at a position closer to an end portion of the core pipe than the first contact member in a case where the core portion is inserted in the core pipe.
8. The printing apparatus according to claim 7, wherein the cam member is able to pivot to the first position in which the first contact member and the second contact member are pressed onto the inner wall and the second position in which the first contact member and the second contact member are retracted from the inner wall.
9. The printing apparatus according to claim 1, wherein the number of the roll holding unit included is two, and the two roll holding units are inserted into the hollow portion of the core pipe from two ends of the core pipe, respectively.
10. The printing apparatus according to claim 1, wherein the contact member has elasticity.
11. A roll holding device that holds a roll in which a sheet is rolled up around a core pipe, comprising: a core portion extending from one side of a flange of a roll holding unit along a rotation axis of the roll and configured to be inserted into a hollow portion of the core pipe from an end portion of the core pipe; a shaft portion extending from the other side of the flange along the

rotation axis; a flange bearing including a bearing and configured to rotatably support the shaft portion by the bearing in a case where the roll holding unit is mounted on the printing apparatus; a contact member provided to the core portion and configured to be able to be put in contact with an inner wall surrounding the hollow portion of the core pipe; and a cam member disposed inside the core portion and configured rotate about the rotation axis between a first position in which the contact member is pressed onto the inner wall and a second position in which the contact member is retracted from the inner wall.